# Electrons on the Move – Some Electrochemistry

Chem 151 Lab

# Specific Redox Definitions

Oxidation - atom loses electrons so oxidation number increases (more positive)

Reduction - atom gains electrons, oxidation number decreases (more negative)



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OIL RIG



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LEO the lion goes GER

# Electrochemistry - The Basics

- Oxidation-Reduction Reactions (Redox) - reactions where electrons are transferred between reagents.
- Oxidation - when an atom or element loses electrons
  - Ex: $Ag(s) \rightarrow Ag^{+} + e^{-}$
- Reduction - when an atom or element gains electrons
  - Ex: $Cu^{+2} + 2e^{-} \rightarrow Cu(s)$

# Redox Reactions

- Oxidation and reduction reactions always occur together!!!
- In overall chemical equation the **electrons** must be balanced.

See Section 7.2 Oxidation Reduction Reactions

# Half Reactions and Notation

For Reduction Half Reactions:

more negative → favors oxidation

more positive → favors reduction

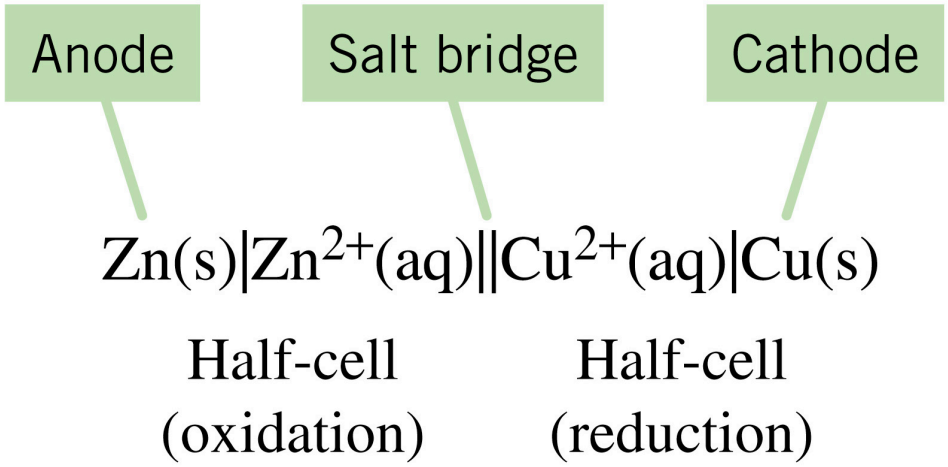



TABLE 25.3 Standard reduction voltages at 25.0°C for aqueous solutions (see also Appendix G)*

| Electrode half reaction | $E^\circ_{red}$/V |
|---|---|
| **Acidic solutions** | |
| $F_2(g) + 2e^- \rightarrow 2F^-(aq)$ | +2.866 |
| $O_3(g) + 2H^+(aq) + 2e^- \rightarrow O_2(g) + H_2O(l)$ | +2.076 |
| $Co^{3+}(aq) + e^- \rightarrow Co^{2+}(aq)$ | +1.92 |
| $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$ | +1.358 |
| $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ | +1.229 |
| $Pt^{2+}(aq) + 2e^- \rightarrow Pt(s)$ | +1.18 |
| $NO_3^-(aq) + 4H^+(aq) + 3e^- \rightarrow NO(g) + 2H_2O(l)$ | +0.957 |
| $Ag^+(aq) + e^- \rightarrow Ag(s)$ | +0.7996 |
| $Cu^+(aq) + e^- \rightarrow Cu(s)$ | +0.521 |
| $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ | +0.342 |
| $Hg_2Cl_2(s) + 2e^- \rightarrow 2Hg(l) + 2Cl^-(aq)$ | +0.268 |
| $AgCl(s) + e^- \rightarrow Ag(s) + Cl^-(aq)$ | +0.2223 |
| $Cu^{2+}(aq) + e^- \rightarrow Cu^+(aq)$ | +0.153 |
| $2H^+(aq) + 2e^- \rightarrow H_2(g)$ | +0.0 |
| $Pb^{2+}(aq) + 2e^- \rightarrow Pb(s)$ | −0.126 |
| $V^{3+}(aq) + e^- \rightarrow V^{2+}(aq)$ | −0.255 |
| $Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$ | −0.447 |
| $Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$ | −0.762 |
| $Mn^{2+}(aq) + 2e^- \rightarrow Mn(s)$ | −1.185 |
| $Al^{3+}(aq) + 3e^- \rightarrow Al(s)$ | −1.662 |
| $H_2(g) + 2e^- \rightarrow 2H^-(aq)$ | −2.23 |
| $Mg^{2+}(aq) + 2e^- \rightarrow Mg(s)$ | −2.372 |
| $Na^+(aq) + e^- \rightarrow Na(s)$ | −2.71 |
| $Ca^{2+}(aq) + 2e^- \rightarrow Ca(s)$ | −2.868 |
| $K^+(aq) + e^- \rightarrow K(s)$ | −2.931 |
| $Li^+(aq) + e^- \rightarrow Li(s)$ | −3.0401 |

← increasing strength of oxidizing agents

increasing strength of reducing agents →

# Electrochemical Potential

- Electrical current is produced by coupling oxid/red chemical reactions.

$$Ag^{+} + e^{-} \rightarrow Ag(s) \quad E = 0.80V$$

$$Cu(s) \rightarrow Cu^{+2} + 2e^{-} \quad E = -0.34V$$

# Standard Electrode Potentials

A **standard electrode potential**, $E^\circ$, is based on the tendency for reduction to occur at the electrode

The cell voltage, called the **standard cell potential** ($E^\circ_{cell}$), is the *difference* of the standard **reduction** potentials of the *cathode* and that of the *anode*

$$E^\circ_{cell} = E_{cathode}^\circ - E_{anode}^\circ$$

$$E^\circ_{cell} > 0 = SPONTANEOUS$$

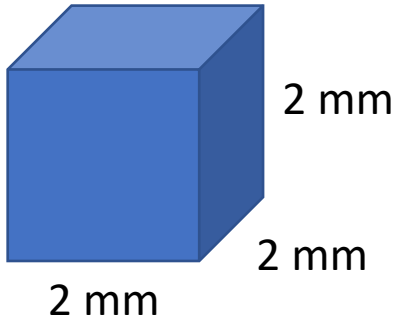
$$E^\circ_{cell} < 0 = NON\ SPONTANEOUS$$

# Procedures:

A. Reactivity
   - what does this look like??

B. Cuperating an Aluminum can

C. Oxidation of Sodium Metal by $H_2O$
   - Do this in the **fume hood**!
     - 8 $mm^3$ means:




SMALL!!!

D. Measuring Potentials

# Equation to Think About:

$$E_{cell} = E^{\circ}_{cell} - \left(\frac{0.02570\ V}{v_e}\right) \ln Q$$

$v_e$ represents the stoichiometric coefficient or the number of electrons transferred in a redox reaction

Q > K means reaction ⟸ $E_{cell}$ goes ??

Q < K means reaction ⟹ $E_{cell}$ goes ??

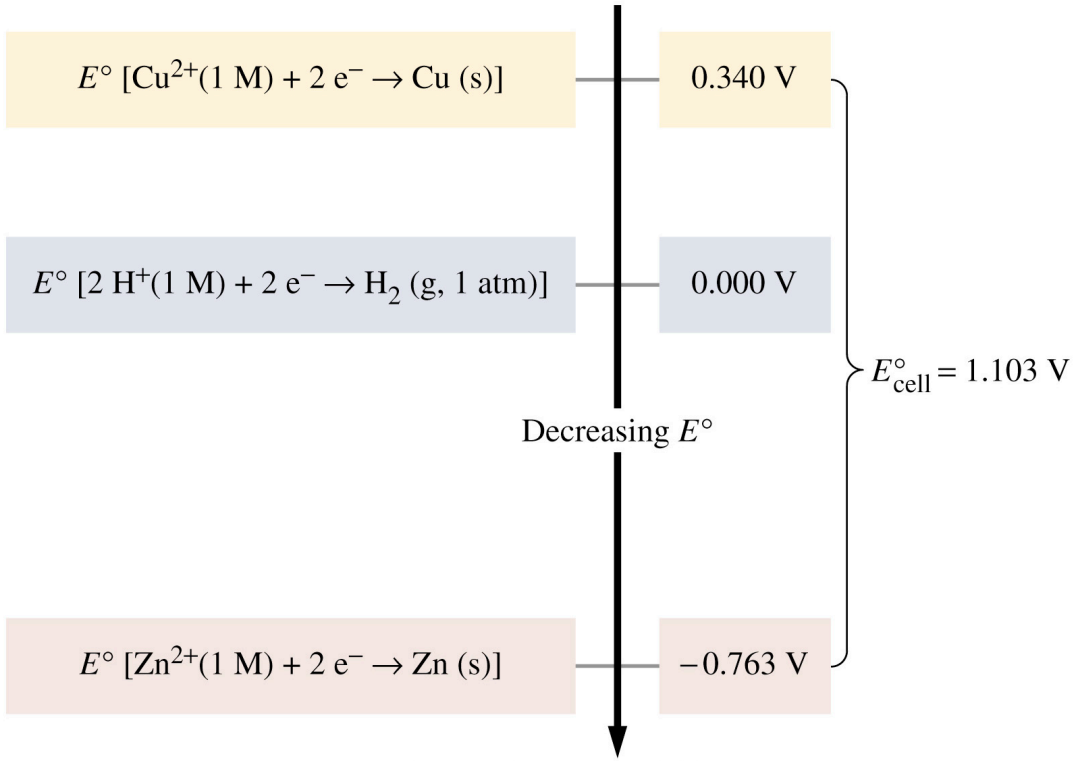
# Part D:

$$Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$$

$$E°_{\text{cell}} = E_{cathode}° - E_{anode}°$$

$$E°_{\text{cell}} > 0 = SPONTANEOUS$$

$$E°_{\text{cell}} < 0 = NON\ SPONTANEOUS$$



# Safety and Clean-up

- Dispose of all waste into the appropriate liquid or solid waste containers in the hood
- Recycle larger Cu and Zn metals
- Be sure to turn in your electronic notebook upload by 11:59PM on Sunday